

Lab - LDAP

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System and Network admin

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Part 1: Configure the LDAP Server (Ubuntu 24.04 LTS)

Perform these steps on the machine designated as the **LDAP Server**.

1. Install OpenLDAP

Update the package list:

```
sudo apt update
```

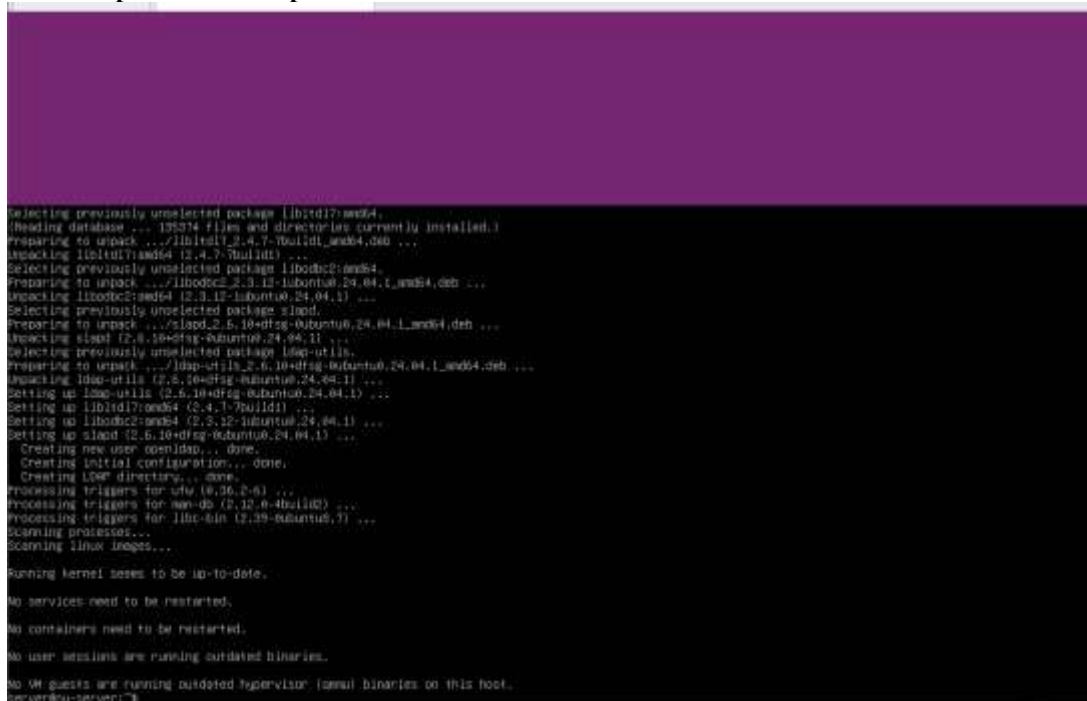
Install the OpenLDAP server and utilities:

```
sudo apt install slapd ldap-utils
```

During installation, you will be prompted to set an administrator password for LDAP.

Choose a secure password and remember it.

 **Screenshot #1:** Capture the terminal output showing the successful completion of the slapd and ldap-utils installation.



```
Selecting previously unselected package libldap2:amd64.  
Reading database ... 175274 files and directories currently installed.)  
Preparing to unpack .../libldap2_2.4.7-7ubuntu1_amd64.deb ...  
Unpacking libldap2:amd64 (2.4.7-7ubuntu1) ...  
Selecting previously unselected package libodbc2:amd64.  
Preparing to unpack .../libodbc2_2.3.12-1ubuntu1_amd64.deb ...  
Unpacking libodbc2:amd64 (2.3.12-1ubuntu1) ...  
Selecting previously unselected package slapd.  
Preparing to unpack .../slapd_2.6.10+dfsg-ubuntu24.04.1_amd64.deb ...  
Unpacking slapd (2.6.10+dfsg-ubuntu24.04.1) ...  
Selecting previously unselected package ldap-utils.  
Preparing to unpack .../ldap-utils_2.6.10+dfsg-ubuntu24.04.1_amd64.deb ...  
Unpacking ldap-utils (2.6.10+dfsg-ubuntu24.04.1) ...  
Setting up libldap2:amd64 (2.4.7-7ubuntu1) ...  
Setting up libodbc2:amd64 (2.3.12-1ubuntu1) ...  
Setting up slapd (2.6.10+dfsg-ubuntu24.04.1) ...  
Creating new user slapd... done.  
Creating initial configuration... done.  
Creating LDAP directory... done.  
Processing triggers for ufw (0.36.2-6) ...  
Processing triggers for man-db (2.12.0-4build1) ...  
Processing triggers for libc-bin (2.39-0ubuntu1) ...  
Scanning processes...  
Scanning linux images...  
  
Running kernel does not need to be up-to-date.  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
server@server:~$
```

2. Configure Base DN

Reconfigure slapd to set up your domain:

```
sudo dpkg-reconfigure slapd
```

Follow the prompts carefully:

- **Omit OpenLDAP server configuration:** No
- **DNS domain name:** Enter your desired domain name (e.g., example.local). This creates your Base DN (e.g., dc=example,dc=local). Confirm it.
- **Organization name:** Enter your organization name (e.g., My [Lab](#) Org).
- **Administrator password:** Enter the password you set during installation. Confirm it.
- **Database backend:** Choose MDB.
- **Remove database when slapd is purged:** No
- **Move old database:** Yes
- **Allow LDAPv2 protocol:** No (LDAPv3 is standard).

 **Screenshot #2:** Capture the final screen of the dpkg-reconfigure slapd process confirming the configuration.

```
Backing up /etc/ldap/slapd.d in /var/backups/slapd-2.6.10+dfsg-0ubuntu0.24.04.1... done.
Moving old database directory to /var/backups:
- directory unknown... done.
Creating initial configuration... done.
Creating LDAP directory... done.
server@ny-server:~$
```

3. Add Basic Structure (Organizational Units)

Create an LDIF file named `base_structure.ldif` using a text editor like nano:

`nano base_structure.ldif`

Add the following content, replacing `dc=example,dc=local` with your actual Base DN configured in the previous step:

`dn: ou=people,dc=example,dc=local`

`objectClass: organizationalUnit`

`ou: people`

`dn: ou=groups,dc=example,dc=local`

`objectClass: organizationalUnit`

`ou: groups`

Save and close the file (Ctrl+X, then Y, then Enter in nano).

Add this structure to the LDAP directory using `ldapadd`.

Replace `cn=admin,dc=example,dc=local` with your admin DN if different.

`sudo ldapadd -x -D cn=admin,dc=example,dc=local -W -f base_structure.ldif`

Enter your LDAP administrator password when prompted.

 **Screenshot #3:** Capture the terminal output showing the successful addition of the two entries (adding new entry "ou=people..." and adding new entry "ou=groups...").

```
server@ny-server:~$ sudo ldapadd -x -D cn=admin,dc=example,dc=local -W -f base_structure.ldif
Enter LDAP Password:
ldap_bind: Invalid credentials (49)
server@ny-server:~$ sudo ldapadd -x -D cn=admin,dc=example,dc=local -W -f base_structure.ldif
Enter LDAP Password:
ldap_bind: Server is unwilling to perform (53)
    additional info: unauthenticated bind (DN with no password) disallowed
server@ny-server:~$
```

4. Add a Test User

Generate an encrypted password for the user:

`slappasswd`

Enter a password (e.g., `testpass123`) twice. Copy the resulting hash (it starts with `{SSHA}`).

Create an LDIF file named `testuser.ldif`:

`nano testuser.ldif`

Add the following content, replacing the placeholder values:

- Replace `dc=example,dc=local` with your Base DN.
- Choose a unique `uidNumber` (e.g., `10001`). Ensure this doesn't conflict with local UIDs on the client.
- Choose a unique `gidNumber` (e.g., `10001`). You might need to create a corresponding group later for full functionality, but this is okay for initial testing.
- Paste the copied password hash after `userPassword:`.

`dn: uid=testuser,ou=people,dc=example,dc=local`

objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
cn: Test User
sn: User
uid: testuser
uidNumber: 10001
gidNumber: 10001
homeDirectory: /home/testuser
loginShell: /bin/bash
userPassword: {SSHA}YourEncryptedPasswordHashHere
Save and close the file.

Add the user to LDAP:

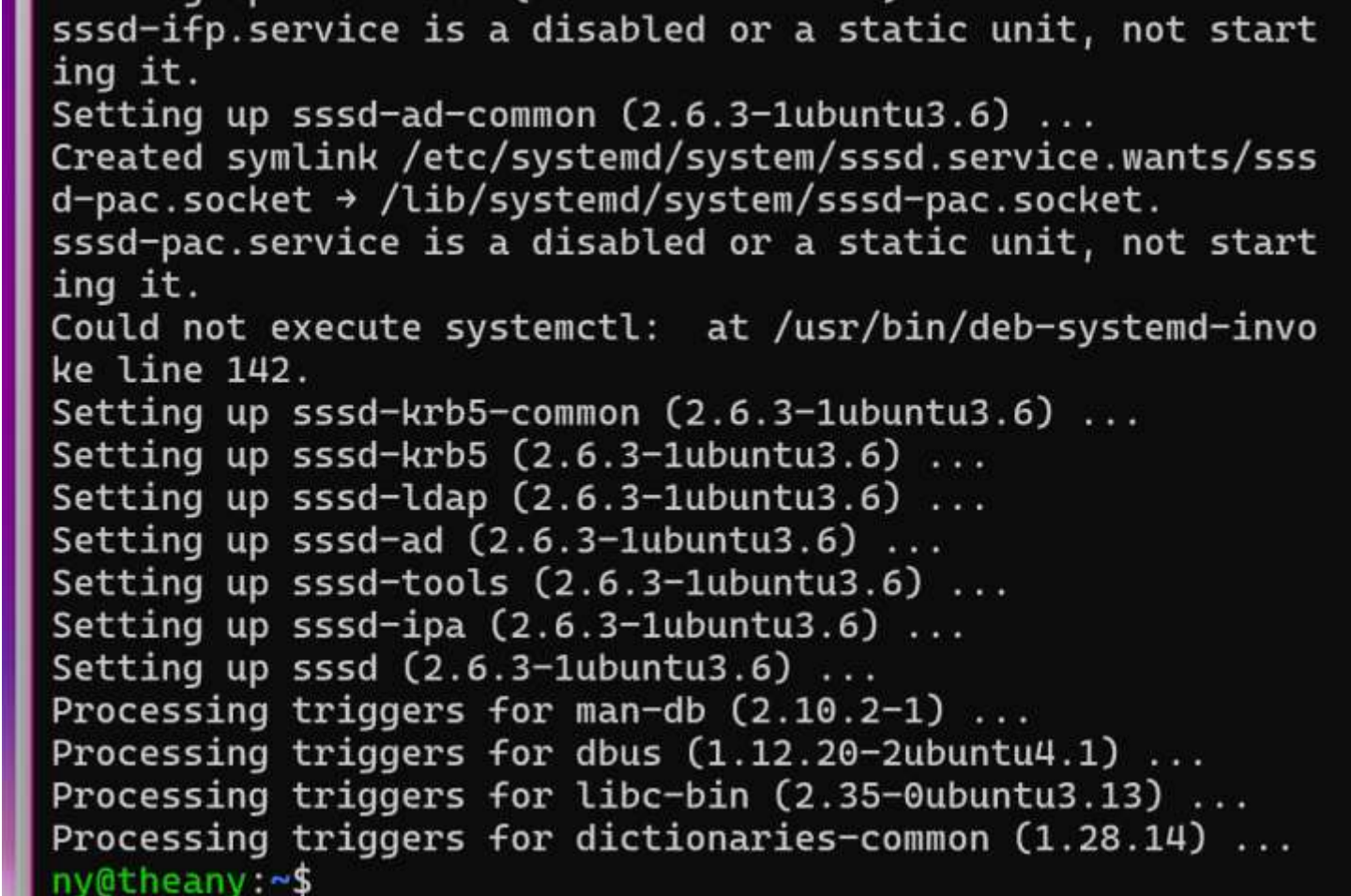
```
sudo ldapadd -x -D cn=admin,dc=example,dc=local -W -f testuser.ldif
```

Enter your LDAP administrator password when prompted.

Verify the user was added by searching:

```
ldapsearch -x -b dc=example,dc=local '(uid=testuser)'
```

 **Screenshot #4:** Capture the terminal output of the ldapsearch command showing the details of testuser.



```
sssd-ifp.service is a disabled or a static unit, not starting it.  
Setting up sssd-ad-common (2.6.3-1ubuntu3.6) ...  
Created symlink /etc/systemd/system/sss.service.wants/sss  
d-pac.socket → /lib/systemd/system/sss-d-pac.socket.  
sssd-pac.service is a disabled or a static unit, not starting it.  
Could not execute systemctl: at /usr/bin/deb-systemd-invo  
ke line 142.  
Setting up sssd-krb5-common (2.6.3-1ubuntu3.6) ...  
Setting up sssd-krb5 (2.6.3-1ubuntu3.6) ...  
Setting up sssd-ldap (2.6.3-1ubuntu3.6) ...  
Setting up sssd-ad (2.6.3-1ubuntu3.6) ...  
Setting up sssd-tools (2.6.3-1ubuntu3.6) ...  
Setting up sssd-ipa (2.6.3-1ubuntu3.6) ...  
Setting up sssd (2.6.3-1ubuntu3.6) ...  
Processing triggers for man-db (2.10.2-1) ...  
Processing triggers for dbus (1.12.20-2ubuntu4.1) ...  
Processing triggers for libc-bin (2.35-0ubuntu3.13) ...  
Processing triggers for dictionaries-common (1.28.14) ...  
ny@theany:~$
```

Part 2: Configure the LDAP Client (Ubuntu 24.04 LTS)

Perform these steps on the machine designated as the **LDAP Client**. Ubuntu 24.04 uses **SSSD** (System Security Services Daemon) as the standard for LDAP authentication.

1. Install LDAP Client Packages (SSSD)

Update the package list:

```
sudo apt update
```

Install the necessary SSSD packages and LDAP utilities:

```
sudo apt install sssd sssd-tools sssd-ldap ldap-utils
```

 **Screenshot #5:** Capture the terminal output showing the successful completion of the client package installation (including sssd).

```
Executing: /lib/systemd/systemd-sysv-install enable sssd
ny@theany:~$ sudo systemctl status sssd
● sssd.service - System Security Services Daemon
   Loaded: loaded (/lib/systemd/system/sss.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2026-05-12 16:20:12 CEST; 1min 15s ago
     Main PID: 2603 (sss)
        Tasks: 4 (limit: 9331)
      Memory: 37.9M
         CPU: 82ms
    CGroup: /system.slice/sss.service
            └─2603 /usr/sbin/sss -i --logger=files
               └─2604 /usr/libexec/sss/sss_be --domain LDAP
                  └─2605 /usr/libexec/sss/sss_nss --uid 0 --
                     └─2606 /usr/libexec/sss/sss_pam --uid 0 --

May 12 16:20:12 theany systemd[1]: Starting System Security Services Daemon: sssd.
May 12 16:20:12 theany sssd[2603]: Starting up
May 12 16:20:12 theany sss_be[2604]: Starting up
May 12 16:20:12 theany sss_nss[2605]: Starting up
May 12 16:20:12 theany sss_pam[2606]: Starting up
May 12 16:20:12 theany systemd[1]: Started System Security Services Daemon: sssd.
```

2. Configure SSSD

Ubuntu 24.04 requires manual creation of the SSSD configuration file. Create it using nano:

```
sudo nano /etc/sss/sss.conf
```

Add the following configuration, making sure to replace

the `ldap_uri` and `ldap_search_base` with your specific server IP/hostname and Base DN:

```
[sss]
```

```
config_file_version = 2
```

```
services = nss, pam
```

```
domains = LDAP
```

```
[domain/LDAP]
```

```
id_provider = ldap
```

```
auth_provider = ldap
```

```
ldap_uri = ldap://<your_ldap_server_ip_or_hostname>
```

```
ldap_search_base = dc=example,dc=local
```



```
ldap_id_use_start_tls = False
ldap_tls_reqcert = never
cache_credentials = True
enumerate = True
```

Save and close the file (Ctrl+X, then Y, then Enter).

Crucial Step: SSSD is extremely strict about permissions. You must change the file permissions to 600 (read/write for root only), or the service will refuse to start:

```
sudo chmod 600 /etc/sss/sss.conf
```

3. Configure Home Directory Creation (PAM)

Configure the system to automatically create a home directory when the LDAP user logs in for the first time. Run:

```
sudo pam-auth-update --enable mkhomedir
```

This will open a dialog. Ensure "Create home directory on login" is checked (marked with an asterisk [*]). Press Tab to select **Ok** and press Enter.

4. Restart and Enable SSSD

Restart the SSSD service to apply the new configuration and enable it to start on boot:

```
sudo systemctl restart sssd
```

```
sudo systemctl enable sssd
```

Check the status to ensure it is running without errors:

```
sudo systemctl status sssd
```

 **Screenshot #6:** Capture the terminal output showing the active (running) status for the sssd service. (Use 'q' to exit the status view if needed).

```
server@ny-server:~$ sudo ldapadd -x -D "cn=admin,dc=example" -W -f testuser.ldif

Enter LDAP Password:
ldap_bind: Server is unwilling to perform (53)
    additional info: unauthenticated bind (DN with no password) disallowed
server@ny-server:~$ sudo ldapadd -x -D "cn=admin,dc=example" -W -f testuser.ldif
Enter LDAP Password:
adding new entry "uid=testuser,dc=example"
ldap_add: Protocol error (2)
    additional info: no attributes provided

server@ny-server:~$
```

5. Test LDAP Login

Verify that the client can see the LDAP user:

```
getent passwd testuser
```

The output should show the details for testuser similar to how it would appear in /etc/passwd.

 **Screenshot #7:** Capture the terminal output of the getent passwd testuser command showing the user's details retrieved from LDAP.

```
server@ny-server:~$ sudo ldapadd -x -D "cn=admin,dc=example" -W -f testuser.ldif
Enter LDAP Password:
adding new entry "uid=testuser,dc=example"
ldap_add: Protocol error (2)
    additional info: no attributes provided

server@ny-server:~$
```

Now, try to switch to the test user:

su - testuser

Enter the password you created for testuser (e.g., testpass123).

If successful, you will be logged in as testuser. You can verify by running whoami.

whoami

pwd

Exit the test user shell:

exit

 **Screenshot #8:** Capture the terminal output showing the successful login using su - testuser, the output of whoami showing testuser, and the output of pwd showing the user's home directory (which was automatically created by PAM).